

HW 6

1) i) The property that is being violated is the integral must be equal to 1. If λ is negative, then the function is $ce^{\lambda x}$ to a positive power, which goes infinitely. The integral would therefore be from $-\infty \rightarrow \infty$, and then diverge. If $\lambda = 0$, then $f(x) = ce^0 = c(1)$. A constant from $-\infty$ to ∞ would also diverge.

$$1.2) \int_{-\infty}^0 ce^{-2\lambda|x|} + \int_0^{\infty} ce^{-10\lambda|x|} = 1$$

The fundamental property of a PDF is that the integrals must sum to 1. Solving the integral we get:

$$\begin{aligned} & c \left(\left[-\frac{1}{2\lambda} e^{-2\lambda|x|} \right]_{-\infty}^0 + \left[-\frac{1}{10\lambda} e^{-10\lambda|x|} \right]_0^{\infty} \right) = 1 \\ & = c \left(\left(-\frac{1}{2\lambda} (e^0 - 0) \right) + \left(-\frac{1}{10\lambda} (0 - e^0) \right) \right) \\ & = c \left(-\frac{1}{2\lambda} + \frac{1}{10\lambda} \right) = 1 \\ & = c \left(\frac{5}{10\lambda} + \frac{1}{10\lambda} \right) = \frac{6c}{10\lambda} = 1 \Rightarrow \frac{3}{5\lambda} c = 1 \end{aligned}$$

$$\frac{5\lambda}{3} = c$$

Because we Assume $\lambda > 0$, We know that when $x < 0$, x is negative, so the -2λ actually acts as positive for our calculations

$$1.3) E[x] = \int_{-\infty}^0 (x \cdot ce^{-2\lambda x}) dx + \int_0^{\infty} (x \cdot ce^{-10\lambda x}) dx$$

$c = \frac{5\lambda}{3}$

$$E[x^2] = \lambda^2 \left(\int_{-\infty}^0 ce^{2\lambda x} dx + \int_0^{\infty} ce^{-10\lambda x} dx \right)$$

$$\begin{aligned} \text{Var}(x) &= (E[x^2]) - (E[x])^2 \\ &= \left(\lambda^2 \left(\int_{-\infty}^0 ce^{2\lambda x} dx + \int_0^{\infty} ce^{-10\lambda x} dx \right) \right) - \left(\lambda \left(\int_{-\infty}^0 ce^{2\lambda x} dx + \int_0^{\infty} ce^{-10\lambda x} dx \right) \right)^2 \\ &= \frac{13}{50\lambda^2} \end{aligned}$$

$$E[x] = -\frac{2}{5\lambda}$$

1.4) Two Cases: $x \geq 0$, $x < 0$ add them together

$$x \geq 0: P(X \geq x) = \int_x^{\infty} \frac{5\lambda}{3} e^{-10\lambda t} dt$$

$$= \left[\frac{5\lambda}{3} \cdot \frac{1}{-10\lambda} e^{-10\lambda t} \right]_x^{\infty}$$

$$= \left[-\frac{1}{6} e^{-10\lambda t} \right]_x^{\infty} = 0 + \frac{1}{6} e^{-10\lambda x}$$

$$x < 0 = \int_x^0 \frac{5\lambda}{3} e^{2\lambda t} dt + \int_0^{\infty} \frac{5\lambda}{3} e^{-10\lambda t} dt$$

When $x < 0$, we need to split the integral, we also know from 1.2, that $\int_0^{\infty} \frac{5\lambda}{3} e^{-10\lambda t} dt = 1/6$.
Therefore:

$$\begin{aligned} \left(\frac{5\lambda}{3} \cdot \frac{1}{2} e^{2\lambda t} \right) \Big|_x^0 &= \frac{5\lambda}{6} e^{2\lambda(0)} - \frac{5\lambda}{6} e^{2\lambda x} \\ &= \frac{5\lambda}{6} - \frac{5\lambda}{6} e^{2\lambda x} \end{aligned}$$

Adding together:

$$\left(\frac{5\lambda}{6} - \frac{5\lambda}{6} e^{2\lambda x} \right) + 1/6 = P(X \geq x)$$

$$\left(\frac{5\lambda}{6} \right) \cdot 1 -$$

$$2) a) f_X(x) = \begin{cases} 0 & x < 0 \\ \frac{x^3}{8b^3} & 0 \leq x \leq 2b \\ 0 & x > 2b \end{cases}$$

We get $\frac{x^3}{8b^3}$ by taking volume of radius x and divide by volume of radius $2b$. Volume is $\frac{4\pi}{3}$ so we get $\frac{\frac{4\pi}{3}x^3}{\frac{4\pi}{3}(2b)^3} = \frac{x^3}{(2b)^3} = \frac{x^3}{8b^3}$

$$b) \text{pdf} = \frac{d}{dx} f_X(x) = \frac{d}{dx} \left(\frac{x^3}{8b^3} \right) = \frac{3x^2}{8b^3}$$

b is a constant, so for d we just do power rule

$$\text{pdf } f_X(x) = \begin{cases} \frac{3x^2}{8b^3} & 0 \leq x \leq 2b \\ 0 & \text{otherwise} \end{cases}$$

$$c) E[X] = \int_0^{2b} x \cdot \frac{3x^2}{8b^3} dx = \frac{3}{8b^3} \int_0^{2b} x^3 dx$$

$$= \frac{3}{8b^3} \left[\frac{1}{4} x^4 \right]_0^{2b} = \frac{3}{8b^3} \left[\frac{16b^4}{4} - 0 \right] = \frac{3 \cdot 4b^4}{8b^3} = \frac{3b}{2}$$

$$2.4) \text{ Variance} = (E(x^2) - E(x)^2)$$

$$E[x^2] = \int_0^a x^2 \cdot \frac{3x^2}{8b^2} = \frac{3}{8b^2} \left[\frac{x^5}{5} \right]_0^{2b} = \frac{3}{8b^2} \cdot \frac{32b^5}{5} = \frac{12b^2}{5}$$

$$\text{Var}(x) = \frac{12b^2}{5} - \left(\frac{3b}{2} \right)^2 = \frac{12b^2}{5} - \frac{9}{4}b^2 = \frac{48}{20}b^2 - \frac{45}{20}b^2 = \frac{3}{20}b^2$$

$$\left(\frac{3}{20}b^2 \right) = \text{var}(x)$$

3) Continuous Uniform RV $\rightarrow$

$$E(x) = \frac{1}{m+1}$$

$$\text{cdf} = f(x) = 1 - (1-x)^m \quad 0 \leq x \leq 1$$

$$\text{pdf} = f(x) = \frac{d}{dx} [1 - (1-x)^m] = m(1-x)^{m-1}$$

$$\text{Var}(x) = (E[x^2] - E(x)^2)$$

$$E(x^2) = \int_0^1 x^2 \cdot f(x) = \int_0^1 x^2 m(1-x)^{m-1} dx$$

$$= \frac{2}{(m+1)(m+2)}$$

$$\text{Var}(x) = \frac{2}{(m+1)(m+2)} - \left(\frac{1}{m+1} \right)^2$$

$$\text{Var}(x) = \frac{2(m+1)}{(m+1)^2(m+2)} - \frac{m+2}{(m+1)^2(m+2)}$$

$$\text{Var}(x) = \frac{2m+2-m-2}{(m+1)^2(m+2)} = \frac{m}{(m+1)^2(m+2)}$$